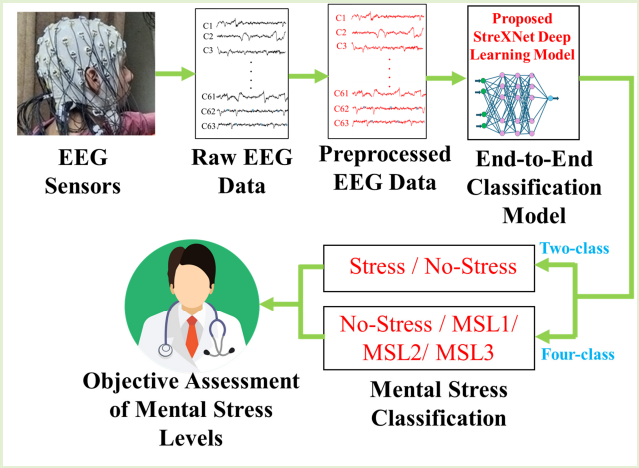
StreXNet: A Novel End-to-End Deep-Learning-Based Improved Multilevel Mental Stress Classification From EEG Sensors

Amit Kumar, *Graduate Student Member, IEEE*, Barath J. K., Shanmukh P., and Deepak Joshi, *Senior Member, IEEE*

***Abstract*—Mental stress has emerged as a psychosocial issue that leads to functional impairment in routine work. It raises the risk of stroke, coronary disease, depression, and hypertension. Evidence suggests a significant impact on cortical brain activity during stressful events. If these brain activities are detected timely, the negative outcomes of stress can be reduced by coping mechanisms, personalized interventions, and timely support. In addition, detection of multilevel stress may facilitate early interventions. Electroencephalography (EEG) sensors have been a potential marker for mental stress classification. However, the heterogeneity in manual feature extraction across researchers makes it difficult to standardize the classification pipeline. Moreover, the feature extraction process requires expert knowledge. To eliminate the need for feature extraction, the present study proposes an end-to-end multilevel mental stress classification from 64-channel EEG sensors using the fusion of extreme gradient boosting (XGBoost), squeeze excitation (SE) block-enhanced convolutional neural network (CNN), and long short-term memory (LSTM) with self-attention-based deep learning model, StreXNet. The results illustrate the improved classification accuracy, in comparison to previous studies, of the proposed StreXNet model with 91.60 ± 3.01% and 91.81 ± 4.26% for two-class and four-class classification, respectively. The one-way repeated measure ANOVA shows a significant difference [$F(4, 68) = 20.261$; $p = 0.000$] and [$F(4, 68) = 46.625$; $p = 0.000$] in classification accuracy between the proposed StreXNet model and other network architectures during two-class and four-class classification, respectively. These findings indicate that the proposed StreXNet deep learning model is more robust and effective for end-to-end classification of multilevel mental stress from EEG sensors.**



***Index Terms*—Deep learning, electroencephalography (EEG) sensors, end-to-end multilevel stress classification, health care industry, mental stress, wearable sensors.**

Received 4 November 2024; accepted 23 November 2024. Date of publication 5 December 2024; date of current version 14 January 2025. The associate editor coordinating the review of this article and approving it for publication was Dr. Tai Fei. *(Corresponding author: Deepak Joshi.)*

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Institute Ethics Committee (IEC) of Indian Institute of Technology (IIT) Delhi under Reference No. 2022/P039.

Amit Kumar and Shanmukh P. are with the Centre for Biomedical Engineering, Indian Institute of Technology Delhi (IIT Delhi), Delhi 110016, India (e-mail: Amit.Kumar@cbme.iitd.ac.in; Shanmukh.34@gmail.com).

Barath J. K. is with the Department of Mechanical Engineering, Columbia University, New York, NY 10027 USA (e-mail: bj2519@columbia.edu).

Deepak Joshi is with the Centre for Biomedical Engineering, IIT Delhi, Delhi 110016, India, and also with the All India Institute of Medical Sciences (AIIMS) Delhi, Delhi 110029, India (e-mail: joshid@cbme.iitd.ac.in).

Digital Object Identifier 10.1109/JSEN.2024.3506984

I. Introduction

MENTAL stress has become a significant concern due to increasing occupational demands, highly competitive work environments, personal responsibilities, and changing lifestyles [1]. The long-term effect of mental stress is one of the most important and common causes of diseases having high mortality rates, such as stroke, cancer, cardiovascular disease, hypertension, and diabetes [2]. Mental stress is a common cause of certain emotions, such as anger, anxiety, schadenfreude, and shame [3]. In addition, the World Health Organization (WHO) recently declared mental stress to be the "Health Epidemic of the 21st Century" [4]. According to the WHO, the long-term effect of mental stress caused by depression affects 350 million individuals worldwide [5]. However, 75% of individuals in low- and middle-income countries never receive treatment due to limited resources for

1558-1748 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.

stress-level detection, insufficient investment in mental health services, a lack of trained healthcare professionals, and the social stigma associated with mental stress [6]. Therefore, a reliable and accurate mental stress classification approach is crucial to address the health challenges associated with mental stress.

In practice, the mental stress is measured through interviews with a qualified psychologist or questionnaires such as the Perceived Stress Scale (PSS) [7], Stress Response Inventory (SRI) [8], Life Events and Coping Inventory (LECI) [9], and State-Trait Anxiety Inventory (STAI) [10]. However, the assessment of these questionnaires is subjective and may differ, especially during multilevel classification of stress, across psychologists [11]. Furthermore, the ignorance of early symptoms by the patients delays the visit to a psychologist, thus delaying the treatment. Therefore, objective evaluation techniques are necessary to ensure an accurate and early diagnosis. Recent neuroscience research indicates significant changes in brain activation during mental stress due to being central to sensory input, emotional responses, and cognitive assessments [12], [13]. Among all modalities available to monitor brain activity during mental stress, electroencephalography (EEG) is the preferred choice due to real-time monitoring in daily life settings (such as home and office) and noninvasive wearable neuroimaging techniques [14], [15], [16].

The mental stress response can be classified as acute or chronic based on the duration of the stressor and the person's adaptive psychological and behavioral coping mechanisms. Acute stress, according to the American Psychological Association, is a short-term physiological response to demands of the recent past or those going to occur in the near future. These can arise from personal losses, nearly missed accidents, arguments with family members, abrupt changes in the environment, or giving a crucial exam [17]. Notably, chronic stress is a long-term condition that arises when people experience acute stress events on a regular basis or when they are under constant pressure due to socioeconomic, interpersonal relationships, or bad careers [18]. Chronic stress is more difficult to mimic in clinical or laboratory settings, and hence acute stress is often addressed in previous research works [19], [20]. In a laboratory setting, acute stress is induced through stimuli such as mental arithmetic task (MAT) [21], [22], [23], Stroop test [24], social stress test (TSST) [25], cold pressor test [26], and public speaking task [27].

Prior to addressing mental stress, it must be quantified into different levels. Quantifying different levels of mental stress is crucial for tailoring personalized interventions and allocating resources effectively, ensuring early support and prevention of more severe mental health issues. Most mental stress classification algorithms rely heavily on techniques of feature selection and extraction and must be recognized by subject-matter specialists [28]. Therefore, an end-to-end approach for multilevel mental stress classification using EEG sensors would be necessary for an objective diagnosis of mental stress levels.

In previous studies, multilevel mental stress classification from EEG sensors has been analyzed through different classification models such as extreme gradient boosting (XGBoost) [29], residual network of 50 layers (ResNet-50) [30], convolutional neural network (CNN), long short-term memory (LSTM) [30], support vector machine (SVM) [16], logistic regression (LR), naive Bayes (NB) [16], linear discriminant analysis (LDA) [31], and random forest (RF) [32]. However, these studies have used manual feature extraction for classification purposes. An end-to-end mental stress classification using CNN [20], symmetric deep convolutional adversarial network (SDCAN), and Euclidean space data alignment approach-based SDCAN (EA-SDCAN) [20] has been studied previously. However, this model lacks the ability to capture both spatial and temporal features of the EEG signal during mental stress classification. In addition, the lack of ability to capture long-range dependencies in the EEG signal during classification has not been considered in previous work.

In summary, the following research gap still exists in existing state-of-the-art (SOTA) literature.

1) Previous existing classification models lack the ability to capture both spatial and temporal features of the EEG signal during mental stress classification.
2) During mental stress classification, previous work has not explored long-range dependence in EEG signals to improve end-to-end classification performance.

To achieve both spatial and temporal dependencies in an end-to-end classification, the present study proposes the fusion of LSTM with self-attention, squeeze excitation (SE) block-enhanced CNN, and an XGBoost classifier-based deep learning model, StreXNet. This hybrid approach leverages LSTM's ability to capture temporal dependencies in EEG data that exhibit dynamic changes over time. By incorporating self-attention mechanisms into LSTM, the model can enhance its focus on relevant parts of the sequence, improving the detection of significant temporal patterns and long-range dependencies. This leads to context-aware temporal feature extraction in the EEG signal. Furthermore, the CNN extracts spatial features from the EEG data, identifying patterns across different channels and brain regions. The integration of XGBoost further refines the spatial and temporal features extracted by the SE-enhanced CNN and LSTM with the self-attention model. XGBoost, with its gradient boosting framework, offers robust handling of the features extracted by LSTM and CNN modules. The combined model benefits from the synergy of these components: LSTM with self-attention for capturing complex temporal dependencies and focusing on important time steps, CNN for detailed spatial feature extraction, and XGBoost for robust classification.

The major contributions of this study are summarized as follows.

1) Instead of the questionnaire-based assessment to estimate the levels of mental stress in an individual, an objective EEG-sensor-based method is used to assess the mental levels.
2) An end-to-end algorithm for multilevel mental stress classification from EEG sensors based on a deep learning model is proposed. The experimental results indicate that this optimized fusion-based deep learning approach improved the accuracy of mental stress classification compared with SOTA deep learning models.

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.

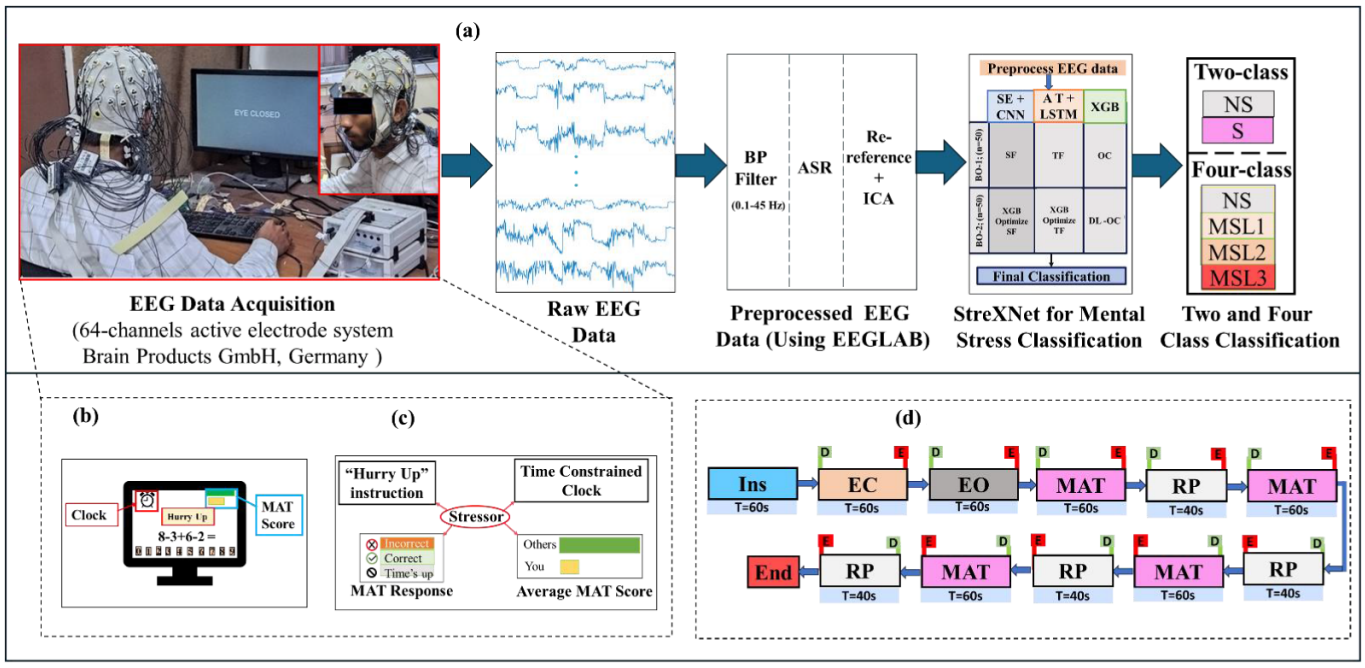


Fig. 1. Overview of end-to-end mental stress classification framework from EEG signal. Here, (a) end-to-end mental stress classification framework from 64-channel EEG signal. BP filter, ASR, and ICA indicate the bandpass filter (0.1–45 Hz), artifact subspace reconstruction, and independent component analysis, respectively. The SE + CNN, AT + LSTM, and XGB represent the squeeze excitation block enhanced convolutional neural network, attention-based long short-term memory, and extreme gradient boosting classifier, respectively. SF, TF, OC, and DL-OC represent the spatial features, temporal features, optimized classifier, and deep-learning-based OC, respectively. BO represents the Bayesian optimization, and "n" represents the number of trials. S, NS, MSL1, MLS2, and MSL3 represent the stress, no-stress, mental stress level 1, mental stress level 2, and mental stress level 3, respectively. (b) Visual instructions on the computer screen. MAT represents the mental arithmetic task. (c) Mental stressor during MAT tasks. (d) Experimental protocol of a representative session during a particular level of MAT. Ins, EC, EO, and RP represent the instruction session, eye-closed condition, eye-open condition, and RP, respectively. "D" and E" psychopy-based automatic trigger during start and end of the block, respectively; and T is the time in seconds.

3) The proposed architecture combines the SE block-enhanced CNN, LSTM with self-attention, and XGBoost classifier to learn spatial and temporal features with long-range dependencies in the EEG signal. The ablation study was performed to observe robust performance during classification of the proposed StreXNet model.

II. Methodology

A. Participants

Eighteen healthy volunteers (mean ± SD, age: 24.78 ± 2.11 years, height: 162.63 ± 3.79 cm, and weight: 67.6 ± 4.36 kg) participated in this experiment. The university-educated individuals were recruited as participants for this study. The international Mini-Mental State Examination (MMSE) was used to assess cognitive and neuropsychological functioning. The MMSE score ranges from 0 to 30. The average (SD) MMSE for the healthy recruited subject was found to be 29.91 ± 0.08. Exclusion criteria included any present or chronic mental or physical illnesses, smoking or substance addiction, and taking medication that might alter EEG measures, which were checked before participation in a standardized interview. In addition, volunteers were instructed to abstain from coffee, food, and exercise for two hours before the experiment. All the participants gave written informed consent before starting the experiment. The study was approved by the Institute Ethics Committee (IEC) of Indian Institute of Technology (IIT) Delhi (Ref. No. 2022/P039). Each participant received a comprehensive detail on the experimental design of the study.

B. Experimental Design

An overview of the mental stress classification framework from the EEG signal is shown in Fig. 1. Fig. 1(a) represents the flow of end-to-end classification of mental stress using a 64-channel EEG sensors system. The preprocessed EEG signal is fed to the proposed StreXNet model for two-class and four-class classification of mental stress. The schematic of visual instructions on the computer screen during the experiment is shown in Fig. 1(b). Similarly, Fig. 1(c) represents the component of the mental stressor to an individual during MAT.

In the present study, the MAT was shown on a computer screen to induce mental stress based on the Montreal Imaging Stress Task (MIST) framework. The MIST was selected because it has demonstrated the ability to produce reliable mental stress that contains the hypothalamus pituitary adrenal (HPA) axis [33]. The MAT was displayed on a computer screen using the Psychopy Builder software [34]. In the present study, MAT was designed with three different difficulty levels. Each MAT level is associated with one particular level of mental stress [16], [21], [35]. At level one, the arithmetic task contains four single-digit integers (ranging between 0 and 9) and uses the operands of "−" and/or "+." The arithmetic task at level two involves four integer numbers, out of which two integers range from 0 to 99, and the remaining two integers range from 0 to 9. This level uses the operands of +, −, and ×. At the third level, the MAT contains four integer numbers, out of which one integer from 0 to 999, two integers ranging from 0 to 99, one integer in a range of 0–9, and the operands include +, −, ×, and ÷. The computer screen displays the

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.

digits "0"–"9" as the available answers for each mental arithmetic question. In this study, we used stress-inducing factors, namely, time constraints and feedback comments associated with each MAT task [22]. For time constraint stressors, each participant received training on the MAT of each difficulty level, and the average time it took each volunteer to complete each arithmetic task was recorded. The participants were put under time pressure conditions by having this recorded time decreased by 10% [22], [36]. The computer screen presented feedback for responses to questions, indicating correctness, incorrectness, or timeouts, along with performance metrics. These metrics included the volunteer's individual performance and the uniformity of average peer performance accuracy, set at 90%. Each participant was given a brief introduction to the proposed tasks.

A schematic representation of the experimental protocol overview with a block design for a given session is shown in Fig. 1(d) and explained as follows: A session started with 60 s of instructions by the experimenter. Next, EEG sensors' data were recorded during the eye close (EC) and eye open (EO) conditions for the next 60 s. Furthermore, the session comprised four (each 60 s) MAT blocks corresponding to a given level of stress condition. Following each 60-s MAT, there was a subsequent 40-s recovery period (RP). The four MAT and subsequent RP were given during the session. Each level of stress had one session. So, in total, three sessions were conducted for each level of stress, namely, mental stress level 1 (MSL1), mental stress level 2 (MSL2), and mental stress level 3 (MSL3). The number of MAT questions in MSL1, MSL2, and MSL3 was 12, 7, and 5, respectively. The EO condition was considered as no-stress (NS) condition during each session. During the 60-s MAT, several arithmetic tasks were presented on the computer screen. During the 40-s RP, volunteers focused their attention on the computer screen's fixation cross. The PsychoPy Builder's code, text, and sound components were used to create the experiment protocols [34]. The Psychopy trigger pulse was used to send the marker as "D" at the start of the block and "E" at the end of each block to label EEG recording. The mean accuracy of responses to questions at each test level was recorded for performance assessment.

C. EEG Data Acquisition

Participants were comfortably seated on a chair nearly 80 cm from the computer screen for the experimental session as [see Fig. 1(a) of EEG Data Acquisition]. EEG data were as captured using a 64-channel Ag/AgCl active electrode system (actiCHAmP, Brain Products GmbH, Germany) positioned at standard scalp locations according to the International 10:10 electrode placement system. A sample rate of 1000 Hz was used to record the EEG data. During recording, channel Fz was used as a reference for EEG data. Under each EEG sensor electrode, conductive gel was used to maintain electrode impedances below 10 kΩ [37].

D. EEG Preprocessing

EEGLAB was used to analyze EEG data collected during the arithmetic task and customized scripts executed in MATLAB (The MathWorks, Academic License IIT Delhi). The data were bandpass filtered between 0.1 and 45 Hz. The 0.1–45-Hz frequency range encompasses several important rhythms, including delta (1–4 Hz), theta (4–8 Hz), alpha (8–14 Hz), beta (14–30 Hz), and low gamma (30–45 Hz) [20], [38]. In addition, the EEG recordings are susceptible to various sources of noise, including power line interference, muscle activity, and eye blinks. Artifact subspace reconstruction (ASR) was used to detect and reconstruct the artifactual segments present in EEG data. ASR analyses EEG data in moving windows using principal component analysis to detect corrupt channels with variance exceeding a set threshold. A moving window of 500-ms duration was used to identify the artifactual portions and standard deviation variance of 20 [39]. The data were then rereferenced using the common average method. The independent component analysis (ICA) algorithm using EEGLAB's "runica" function was used to eliminate artifacts such as eye blinking, eye movement, and muscle activity [40]. The processed EEG data were downsampled to 200 Hz. The z-score normalization was applied to minimize the fluctuation and nonstationarity of the EEG data over time for each channel, as shown in the following equation:

$$\hat{x}_{t,n} = \frac{x_{t,n} - \overline{x}_n}{\sigma_n} \tag{1}$$

where t and n represent the sampling time-point and the channel of the sample, respectively. $\hat{x}_{t,n}$ depicts the normalized value of the original sample $x_{t,n}$. $\overline{x}_n$ and σ_n represent the mean and standard deviation of the sample on channel n, respectively. The average of the EEG data for each channel of the sample after normalization is 0, and its standard deviation is 1.

E. MAT Performance Analysis

The response to an arithmetic question during MAT was recorded for every participant. The MAT performance accuracy (MPA) was further calculated using correct and incorrect responses of arithmetic questions as per (2). The performance accuracy was calculated for all three levels of mental stress (MSL1, MSL2, and MSL3)

$$\text{MPA}\ (\%) = \frac{\text{TCR}}{\text{TNQ}} \times 100. \tag{2}$$

Here, MPA, TCR, and TNQ represent the MPA, total correct response during MAT, and total number of questions during MAT, respectively.

F. Network Architecture

The overall schematic of the deep learning architecture for multilevel mental stress classification from EEG sensors is shown in Fig. 2. The proposed StreXNet structure in this work combines SE block-enhanced CNN, attention-based LSTM (AT-LSTM), and XGBoost to ensure comprehensive feature extraction and robust classification.

1) CNN Block: The CNN block consists of three 1-D convolutional layers, each specifically tailored to extract spatial features from the EEG data. The three convolutional layers use a kernel size of 8, 5, and 3 for the first, second, and third layers,

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.

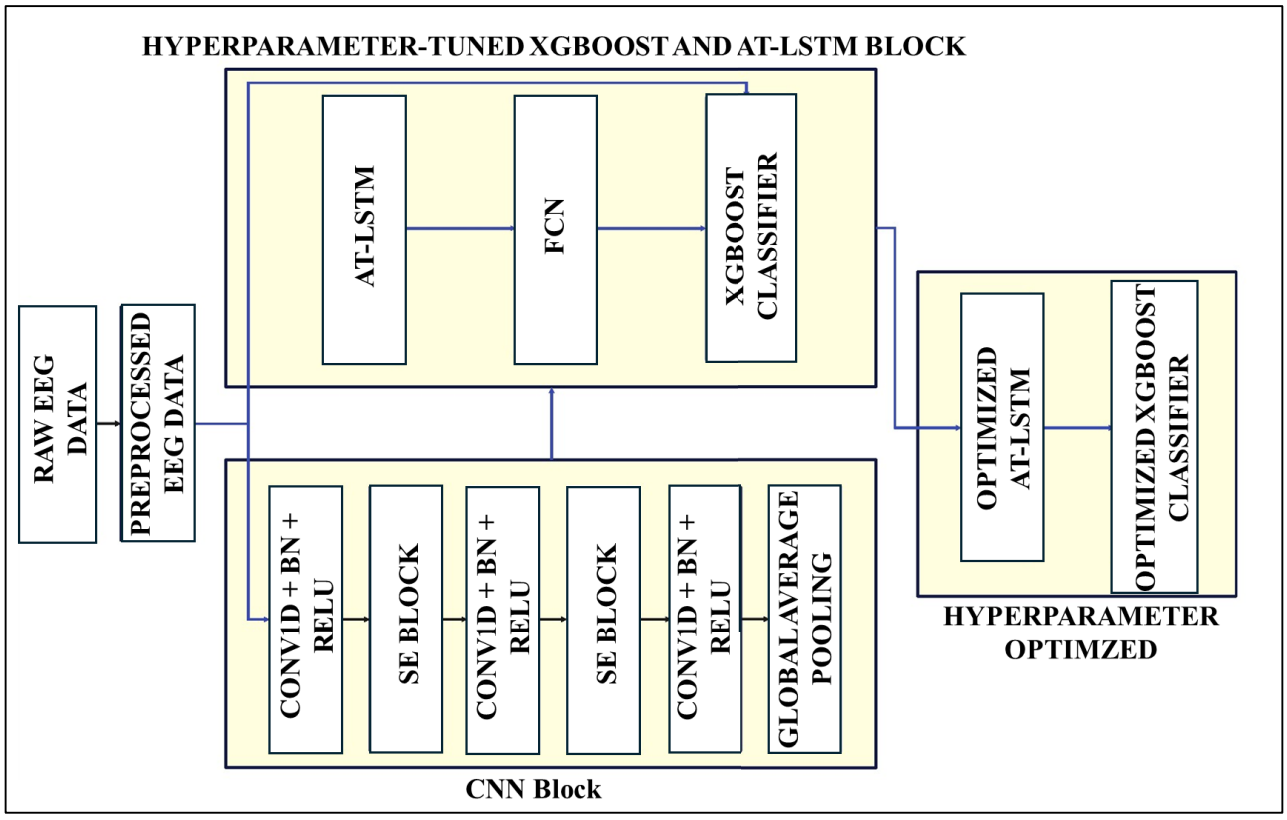


Fig. 2. Schematic of the proposed StreXNet architecture for mental stress classification. In the convolution neural network (CNN) block, Conv1D, BN, ReLU, and SE represent the 1-D convolutional layer, batch normalization layer, rectified linear unit, and squeeze excitation block, respectively. XGBOOST, AT-LSTM, and FCN represent extreme gradient boosting, attention-based long short-term memory, and fully connected networks, respectively.

respectively, to capture the patterns in the EEG data. The CNN block has three batch normalization (BN) layers, following each of the convolutional layers, to stabilize the training process and improve overall model performance [41]. The two SE blocks, positioned after the first and second convolutional layers, enhance the network's ability to focus on the most informative features [42]. Following the convolutional layers, a global average pooling layer is applied to reduce the spatial dimensions of the feature maps and effectively condense the spatial information into a compact vector. Three rectified linear unit (ReLU) activation functions after each convolutional layer are specifically chosen to introduce necessary nonlinearity and prevent the vanishing gradient problem [43]. In addition, using the sigmoid function in SE blocks, the model ensures that feature recalibration enhances overall feature representation and helps in improving classification performance [44].

2) AT-LSTM Block: The proposed StreXNet architecture incorporates an AT-LSTM module to capture temporal dependencies in the EEG signals. This module starts with an LSTM network that is designed to model sequential data, making it particularly effective for handling the time-dependent nature of EEG signals [45]. The LSTM network is initialized with a hidden size that is optimized through Bayesian-optimization-based tuning of hyperparameter for ($N = 50$) trials to find the optimal size for the LSTM's hidden state that balances the model's capacity to learn from the data while avoiding overfitting. The output of the LSTM is then passed through a self-attention mechanism, which assigns varying levels of importance to different time steps. This attention mechanism produces a context vector that encapsulates the most relevant temporal features within the EEG data that are most indicative of mental stress levels. A dropout layer is applied to the context vector generated by the AT-LSTM module to prevent overfitting. The use of a single dropout layer strikes a balance between regularization and model complexity [46].

3) Feature Integration and Final Classification: The fully connected (FC) layer is a crucial component of the proposed model, responsible for integrating the spatial features extracted by the CNN block with the temporal features captured by the AT-LSTM module. This layer combines the context vector from the AT-LSTM with the feature vector produced by the CNN layers, creating a comprehensive, high-dimensional representation of the EEG signals. The FC layer is designed with its architecture and parameters, such as LSTM hidden size, the number of convolutional channels, and the dropout rate optimized through Bayesian optimization to maximize classification performance for ($N = 50$) trials.

Finally, the high-dimensional feature vectors generated by the FC layer are used to train an XGBoost model. XGBoost, known for its robustness and efficiency in handling structured data, is optimized for this task through parallel hyperparameter tuning for each subject. The Bayesian optimization is performed for subject-specific hyperparameter tuning for XGBoost [47]. Key parameters tuned include maximum depth of tree, minimum sum of instance weights, minimum loss reduction, subsamples for fitting, fraction of tree features used,

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.

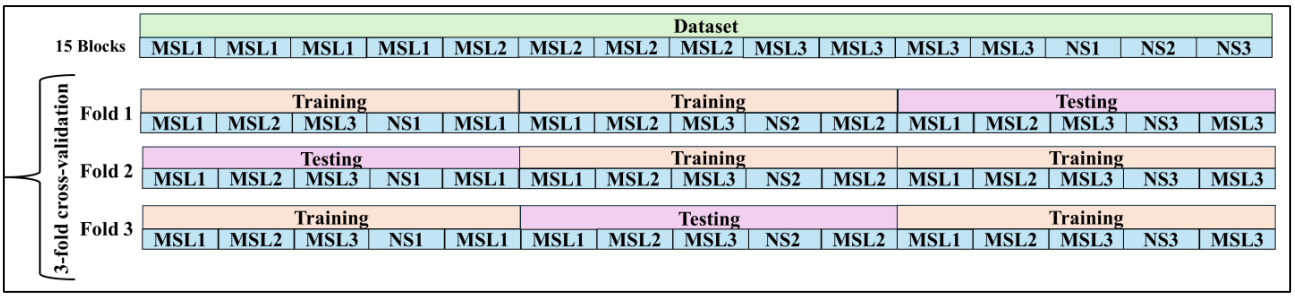


Fig. 3. Schematic of the threefold cross-validation during training and testing for classification models. Here, MSL1, MSL2, and MSL3 blocks represent the EEG data during MAT at mental stress level 1, mental stress level 2, and mental stress level 3, respectively. NS1, NS2, and NS3 block represent EEG data during EO condition as NS from MSL1, MSL2, and MSL3 sessions, respectively.

and L1 regularization terms for ($N = 50$) trials. By combining deep-learning-based feature extraction with XGBoost for final classification, the proposed StreXNet architecture leverages the strengths of both the methodologies.

G. Simultaneous Hyperparameter Optimization

The simultaneous hyperparameter optimization (SHPO) technique developed in this architecture occurs in two stages, focusing on optimizing both the neural network (CNN and AT-LSTM) and the classifier (XGBoost). The novelty of this approach is integrating an optimized XGBoost model within the optimization loop of a deep learning model. Specifically, the deep learning model's hyperparameters, such as the LSTM hidden size (8–64 units), first convolutional layer (32–128), second convolutional layer (64–256), third convolutional layer (32–128), and dropout rate (0.1–0.5), are optimized not just for standalone performance, but for its ability to produce features that significantly enhance the performance of the optimized XGBoost classifier. In this process, parameters of XGBoost, such as tree depth (ranging from 4 to 10), minimum child weight (from 1 to 5), and regularization (from 0.00206 to 0.447) are optimized through a grid search to identify the best-performing hyperparameters. These parameters are then used during the neural network's optimization phase. The CNN and Attention-LSTM network are trained to extract features from EEG data, and these features are passed to the XGBoost model, whose accuracy is used as the evaluation metric during the neural network's hyperparameter tuning. This SHPO, subject-specific optimization results in a highly personalized and effective model that leverages the strengths of both deep learning and traditional machine learning.

H. Performance Evaluation

Subject-specific training and testing were conducted on the dataset using a threefold cross-validation approach. For each subject, four MAT blocks and one block of no stress (EO rest state) [Fig. 1(d)] for each stress level were recorded. Therefore, in total, 12 blocks of MAT (four for each level—MSL1, MSL2, and MSL3) and three blocks of NS levels were recorded as shown in Fig. 3. The EEG data in MAT block and NS were represented by segmenting epochs of 3 s, with an 80% overlap of the epochs. The schematic for threefold cross-validation during training and testing for classification models is shown in Fig. 3. A total of 15 blocks (12 MAT blocks + three no stress blocks) underwent through threefold cross-validation, in which ten blocks were used for training purposes, and the remaining five blocks were used for testing purpose during each fold. The data leakage is effectively prevented as all MAT blocks were separated by an RP/different trail of mental stress level. To improve model training on imbalanced data (only three blocks of MAT in contrast to four blocks of each mental stress level), a weighted cross-entropy loss function was used as per (3). We decided to penalize the classification by imposing additional cost on the model for making classification mistakes in predicting the minority class during training. Thus, each training process aims to minimize the weighted cross-entropy loss, which has been a widely used method in class imbalanced learning problems [48].This ensures that the model pays more attention to the underrepresented classes during training, leading to better generalization and improved classification performance

$$\text{LOSS} = -\frac{1}{K}\sum_{n=1}^{K}\sum_{i=1}^{M} w_i\, p_{ni} \ln(y_{ni}) \tag{3}$$

where K = total number of samples, M = number of classes, w_i = weight for class i, $p_{ni} = 1$ if sample n belongs to class i, 0 otherwise, and y_{ni} = predicted probability that sample n belongs to class i.

In the two-class classification approach, one class comprises all instances where stress levels are not explicitly differentiated (i.e., combining MSL1, MSL2, and MSL3 into a single class), while the other class is dedicated to instances from the NS phase. In the four-class classification scenario, stress levels are classified independently, distinguishing between NS, MSL1, MSL2, and MSL3.

To assess the model's performance in a subject-independent context, we used a leave-one-subject-out cross-validation (LOOCV) approach. In this method, the data from one subject were set aside for testing, while the model was trained on data from the other 17 subjects. This process was repeated 18 times, allowing each subject's data to serve as the test set exactly once. This approach provided a robust evaluation of the model's ability to generalize across different subjects.

The performance during classification was assessed using the following equations:

$$\text{Accuracy} = \frac{T_p + T_N}{T_p + T_N + F_p + F_N} \times 100 \tag{4}$$

$$\text{Precision} = \frac{T_p}{T_p + F_p} \times 100 \tag{5}$$

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.

$$\text{Recall} = \frac{T_p}{T_p + F_N} \times 100 \tag{6}$$

$$F1\text{-score} = \frac{T_p}{T_p + \left(F_p + F_N\right) \times \frac{1}{2}} \times 100 \tag{7}$$

$$\text{MCC} = \frac{(T_p \times T_N) - (F_p \times F_N)}{\sqrt{(T_p + F_p) \times (T_p + F_N) \times (T_N + F_p) \times (T_N + F_N)}} \times 100 \tag{8}$$

$$\text{Jaccard Index (JI)} = \frac{T_p}{T_p + F_p + F_N} \times 100 \tag{9}$$

$$\text{Cohen's Kappa} = \frac{P_o - P_e}{1 - P_e} \times 100 \tag{10}$$

where T_p, T_N, F_p, and F_N indicate the number of true positives, true negatives, false positives, and false negatives, respectively. MCC represents Matthew's correlation coefficient. P_o and P_e represent the actual observed agreements and chance agreements, respectively.

I. Statistical Analysis

The statistical analysis of mental stress classification accuracy was performed for different classification models for two classes (S and NS) and four classes (NS, MSL1, MSL2, and MSL3). A one-way repeated measure analysis of variance (RMANOVA) across the classification models such as RF, AdaBoost, SVM, EEGNet, and the proposed StreXNet was performed to find the significant difference for classification accuracy during two-class and four-class classification. A *post hoc* pairwise analysis was also performed using Bonferroni correction for pairwise comparisons between classification models. Similarly, a one-way RMANOVA across the classification models such as XGBoost, CNN, CNN–LSTM, and XGBoost–CNN, and StreXNet was conducted for classification accuracy during the ablation study in two-class and four-class classification. A *post hoc* pairwise analysis was also performed using Bonferroni correction for pairwise comparisons between classification models. In addition, the statistical analysis of MPA was conducted for difficulty levels of MAT as a factor using one-way RMANOVA. Furthermore, the statistical analysis (using RMANOVA) was performed on precision, recall, $F1$-score, MCC, Jaccard index, and Cohen's kappa across the classification models for two-class and four-class classification. The level of statistical significance was set for all variables at $p \leq 0.05$. Statistical analysis was performed using SPSS (version 20, SPSS Inc., Chicago, IL, USA).

III. Results

In this section, a subject-dependent analysis is presented to assess the efficacy of the proposed novel deep neural network architecture (StreXNet) for multilevel mental stress classification from EEG sensors. A detailed comparison of performance parameters between the StreXNet model and the other contemporary models is also presented.

A. Experimentation Details

StreXNet is implemented using the PyTorch library in Python with a V100 GPU (13-GB RAM). The system used the NVIDIA CUDA Compiler version 12.2.140. The machine model used is a Google Compute Engine which uses Ubuntu 22.04.3 LTS (Jammy Jellyfish) and runs on Intel Xeon single core 2.2 GHz, 64 bits with L2 cache of 256 kiB. The model was trained using an Adam optimizer. The training process involved running the model through 50 epochs while maintaining a learning rate of 0.001.

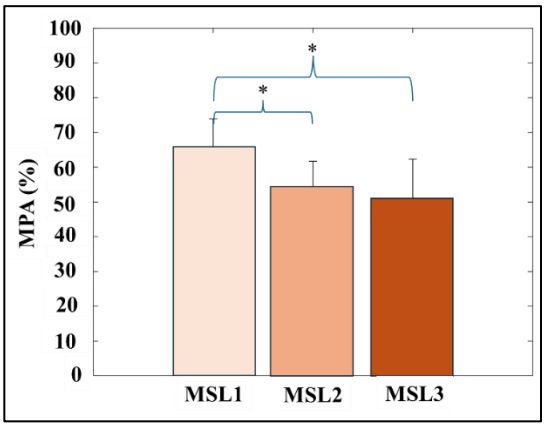


Fig. 4. Statistical results of MPA for three difficulty levels. Here, MSL1, MSL2, and MSL3 represent mental stress level 1, mental stress level 2, and mental stress level 3, respectively. The asterisk (*) indicates the significant difference with $p < 0.05$.

B. MAT Performance

The present study includes three different mental stress levels (MSL1, MSL2, and MSL3) using MAT. The MPA was recorded for each individual at three levels during the MAT response. The mean (SD) of MPA (%) across the subjects was found to be 65.86 ± 8.02%, 54.40 ± 7.35%, and 50.93 ± 11.39% for MSL1, MSL2, and MSL3, respectively, as shown in Fig. 4. The statistical analysis (one-way repeated measure ANOVA) showed a significant effect [$F(2, 34) = 38.488$; $p = 0.000$] was observed for MPA between mental stress levels through MAT. Furthermore, a pairwise comparison (using Bonferroni correction) found a significant difference in MAT performance between MSL1 and MSL2 ($p = 0.000$), and MSL3 and MSL1 ($p = 0.000$). However, no significant difference was found between MSL2 and MSL3 ($p = 0.289$).

C. Classification Performance (Two-Class Versus Four Class)

Table I represents the classification performance of RF, AdaBoost, SVM, EEGNet, and the proposed StreXNet model for two-class and four-class classification. The RF model has achieved an average (across subjects) accuracy of 83.74% in two-class classification and 66.11% in four-class classification.

However, its recall, particularly in two-class classification, was concerning at 60.31%, indicating limitations in capturing the complex patterns inherent in EEG data. AdaBoost model has achieved an average accuracy of 82.84% in two-class classification and 52.05% in four-class classification. Furthermore, SVM is effective in high-dimensional spaces but still struggles with the nonlinear relationships within the data, and it has achieved an average accuracy (across the subjects) of 82.61% and 54.82% for two-class and four-class classification, respectively. EEGNet, designed with depthwise separable

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.

TABLE I
MENTAL STRESS CLASSIFICATION PERFORMANCE AVERAGED ACROSS SUBJECTS ($N = 18$) FOR DIFFERENT CLASSIFICATION MODELS

Models	Two-Class Classification				Four-Class Classification			
	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
RF	83.74±3.48	84.14±10.54	60.31±8.31	61.08±11.31	66.10±6.84	62.66±6.83	64.20±6.10	59.78±5.60
AdaBoost	82.84±3.76	68.28±18.88	60.12±10.73	59.52±13.41	52.05±4.74	54.64±7.53	50.97±4.96	49.08±6.45
SVM	82.61±4.38	60.79±21.43	58.37±12.38	56.12±15.39	54.82±18.69	57.00±16.34	53.44±18.35	51.26±19.28
EEGNet	80.89±9.50	67.73±17.81	66.57±17.31	64.54±17.74	65.07±18.20	62.85±17.03	63.53±16.71	59.43±18.73
Proposed StreXNet	**91.60±3.01**	**87.49±5.10**	**84.88±5.97**	**85.76±5.64**	**91.81±4.26**	**91.57±4.38**	**91.40±4.52**	**91.41±4.49**

Here, RF, AdaBoost, SVM, and EEGNet represent the random forest, adaptive Boosting, support vector machine, and electroencephalography network, respectively.

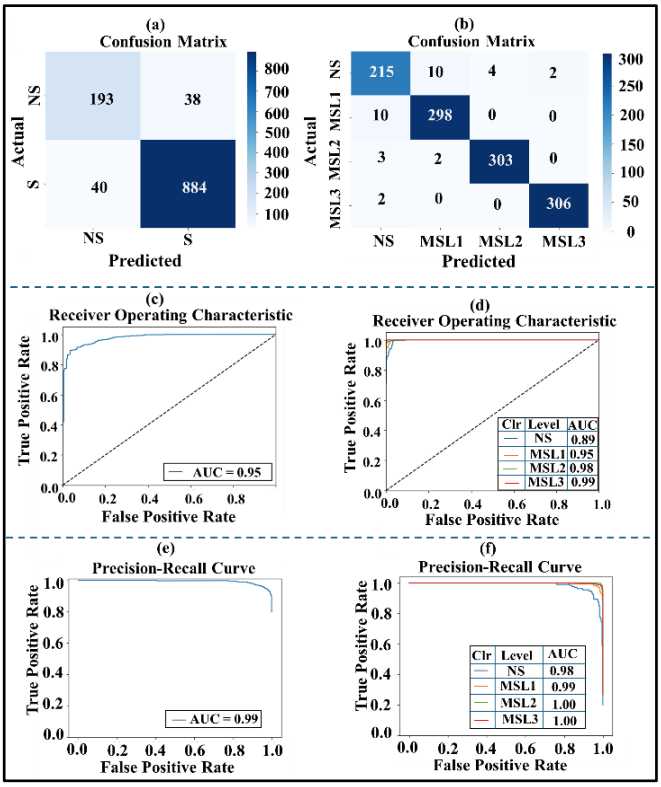


Fig. 5. CM, ROC curve, and PR curve for two-class and four-class mental stress classification from the proposed StreXNet model. (a) CM for two-class classification, (b) CM for four-class classification, (c) ROC curve for two-class classification, (d) ROC curve for four-class classification, (e) PR curve for two-class classification, and (f) PR curve for four-class classification. AUC, Clr, S, NS, MSL1, MSL2, and MSL3 denote the values of area under the curve, color, stress, NS, mental stress level 1, mental stress level 2, and mental stress level 3, respectively.

convolutions, achieved an average accuracy of 80.89% in two-class classification and 65.07% in four-class classification. While each model has its strengths, none fully meets the demands of EEG classification, necessitating further refinement and exploration.

The proposed StreXNet model has achieved an average accuracy of 91.60% and 91.81% for two-class and four-class classification, respectively (see Table I). Fig. 5 represents a representative confusion matrix (CM), receiver operating characteristic (ROC) curve, and precision–recall (PR) curve for a subject for two-class and four-class mental stress classification from the proposed StreXNet model. The model exhibits high accuracy for correct prediction across all classes, 93.1% for NS, 96.8% for MSL1, 98.4% for MSL2, and 99.4% for MSL3 [see Fig. 5(b)]. Furthermore, the multiclass ROC curve indicates that the AUC values for NS (0.89), NS (0.89), MSL1 (0.95), MSL2 (0.98), and MSL3 (0.99) [see Fig. 5(d)]. The AUC values suggest that the multiclass model is highly effective at distinguishing between classes from EEG data. Similarly, the multiclass PR curve displays individual AUC values for NS (0.98), MSL1 (0.99), MSL2 (1.00), and MSL3 (1.00) [see Fig. 5(f)]. These values indicate that the proposed StreXNet model handles multiclass classification with improved precision and recall performance.

The one-way repeated measure ANOVA (across the classification models) shows a significant difference [$F(4, 68) = 13.205$; $p = 0.000$] classification. Furthermore, the pairwise comparison (using Bonferroni correction) found a significant difference between the proposed StreXNet model and other network architectures, such as RF ($p = 0.000$), AdaBoost ($p = 0.000$), SVM ($p = 0.000$), and EEGNet ($p = 0.003$). Similarly, one-way repeated measure ANOVA (across the classification models) shows a significant difference [$F(4, 68) = 27.267$; $p = 0.000$] in classification accuracy during four-class classification. Furthermore, the pairwise comparison (using Bonferroni correction) found a significantly higher accuracy of the proposed StreXNet model in comparison to other network architectures, such as RF ($p = 0.000$), AdaBoost ($p = 0.000$), SVM ($p = 0.000$), and EEGNet ($p = 0.000$).

D. Ablation Study

The development of the proposed StreXNet was a systematic process, with each stage of the training contributing to the refinement of the model's architecture and parameters. This ablation study outlines the sequence of techniques explored, the challenges encountered, and the innovations that ultimately led to the novel classification model from EEG sensors with improved accuracy and performance. The classification performance of the proposed StreXNet model's components, such as XGBoost, CNN, CNN–LSTM, and XGBoost–CNN, was conducted, and the results of the ablation experiment are summarized in Table II.

1) Extreme Gradient Boosting: The XGBoost was chosen for its superior handling of structured data, particularly its ability to manage sparse data and its in-built regularization

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.

TABLE II
MENTAL STRESS CLASSIFICATION PERFORMANCE AVERAGED ACROSS SUBJECTS ($N = 18$) FOR ABLATION STUDY

Models	Two-Class Classification				Four-Class Classification			
	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
XGBoost	85.31±4.16	88.72±5.35	64.01±10.66	65.50±13.61	79.65±5.54	78.19±8.18	77.39±6.20	75.47±7.85
CNN	77.87±8.02	58.93±16.65	57.80±13.71	56.94±14.80	53.12±12.56	52.15±10.99	53.20±11.60	49.59±12.10
CNN - LSTM	80.36±8.41	62.36±16.88	60.82±14.34	59.75±14.75	66.88±14.20	62.04±15.71	64.93±13.07	59.44±15.64
XGBoost-CNN	82.32±2.07	82.97±2.15	98.45±1.07	89.92±1.07	67.08±8.33	64.95±8.62	64.87±7.84	63.33±7.77
Proposed StreXNet	**91.60±3.01**	**87.49±5.10**	**84.88±5.97**	**85.76±5.64**	**91.81±4.26**	**91.57±4.38**	**91.40±4.52**	**91.41±4.49**

Here, XGBoost, CNN, CNN-LSTM, and XGBoost-LSTM represent extreme gradient boosting, convolutional neural network, convolutional neural network-long short-term memory, and extreme gradient boosting-long short-term memory.

mechanisms. Transitioning to XGBoost led to notable improvements, particularly in multiclass classification, where the $F1$-score improved to 75.47%. However, the recall for two-class classification, although improved (64.01%), still did not meet expectations, suggesting that further refinement was necessary. One key reason XGBoost struggled was its reliance on a fixed, tree-based architecture, which has limitations in capturing complex and nonlinear relationships in the data.

2) Convolutional Neural Network: The CNN module-based architecture was chosen next based on the promising results from EEGNet. After testing convolutional layers ranging from one to four, we observed that when the number of convolutional layers (m) exceeded three, the classification metrics fell below those achieved by EEGNet. For configurations with $1 \leq m \leq 3$, specifically when $m = 1$ and $m = 2$, the classification metrics were also unsatisfactory compared with EEGNet, despite selecting this optimal configuration and conducting Bayesian-optimization-based hyperparameter tuning ($N = 50$ trials). Consequently, $m = 3$ was identified as the optimal number of convolutional layers. It begins with an input layer processing EEG data with a shape of [batch size (B) = 32, 63, 600]. Then first, second, and third convolution layers with sizes of (B, 32, 600), (B, 64, 600), and (B, 128, 600), respectively, were used.

3) Fusion of CNN and LSTM: In this architecture, the CNN module begins with processed EEG data with a shape of (B, 63, 600). The CNN module includes three convolutional layers: the first applies 32 filters with a kernel size of 1×8, followed by BN and ReLU activation, resulting in an output shape of (B, 32, 600). The second layer applies 64 filters with a kernel size of 1×5, again followed by BN and ReLU, producing an output shape of (B, 64, 600). The third layer uses 128 filters with a kernel size of 1×3, followed by BN and global average pooling, leading to an output shape of (B, 128, 1). The output from the CNN module is then concatenated with the output from an LSTM module, which captures the temporal dependencies in EEG data. The LSTM processes the input data, extracting sequential patterns that complement the spatial features obtained by the CNN. The combined feature vector is passed through an FC layer with 128 neurons, followed by an output layer with four classes, where a softmax activation function is applied to generate class probabilities. This CNN–LSTM fusion model demonstrated better performance compared WITH standalone CNNs, particularly in two-class classification, where the $F1$-score improved to 59.75%. However, in multiclass classification, the improvement was moderate, with an $F1$-score of 59.44%. This outcome suggested that while the addition of LSTM with hidden size between 8 and 64 and tuned based on objective criterion function helped in capturing sequential patterns, the model still faced challenges in effectively managing the complexity and variability in EEG data.

4) Fusion of XGBoost and CNN: In this architecture, the CNN module with $m = 3$ starts by processing EEG data with a shape of (B, 63, 600). The CNN module includes three convolutional layers: the first applies 32 filters with a kernel size of 1×8, followed by BN and ReLU activation, resulting in an output shape of (B, 32, 600). The second layer applies 64 filters with a kernel size of 1×5, again followed by BN and ReLU, producing an output shape of (B, 64, 600). The third layer uses 128 filters with a kernel size of 1×3, followed by BN and global average pooling, leading to an output shape of (B, 128, 1). The feature maps produced by CNN are then flattened and passed to an XGBoost classifier. The CNN–XGBoost model showed a significant improvement in performance, particularly in two-class classification, where it achieved a recall of 98.46% and an $F1$-score of 89.92%, demonstrating its effectiveness in leveraging the strengths of both THE architectures. However, the performance in multiclass classification remained less satisfactory, with an $F1$-score of 63.32%. This suggests that the CNN–XGBoost combination excels in two-class classification performance. However, it still faces challenges in managing the greater complexity and variability associated with the four-class classification of mental stress using EEG sensors.

5) Fusion of XGBoost, CNN, and LSTM (StreXNet): This architecture was specifically designed to overcome the limitations of previous models by enhancing feature representation, improving the capture of temporal patterns, and focusing on relevant features. The CNN module, for instance, uses 32 filters in the initial convolutional layer with a kernel size of 8, followed by layers with 128 filters and kernel sizes of 5 and 3. These layers are augmented by SE blocks, which operate with a reduction ratio of 16 to adaptively recalibrate channelwise feature responses, ensuring that the model emphasizes the most critical features. The LSTM

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.

component, equipped with a hidden size ranging from 8 to 64 units, is designed to capture complex temporal dependencies within the EEG data. This is further enhanced by a self-attention mechanism, which allows the model to focus on the most relevant temporal features, generating a context vector that encapsulates essential temporal patterns. After this feature extraction phase, the resulting features are fed into an XGBoost classifier. This classifier is pretuned with hyperparameters such as tree depth (4–10), minimum child weight (1–5), and gamma (0.00206–0.447), optimizing it to handle the sophisticated feature space generated by the neural network components. The SHPO technique plays a critical role in this StreXNet model by optimizing the deep learning components (CNN and LSTM) and the XGBoost classifier in a coordinated manner. For instance, hyperparameters such as LSTM hidden size (8–64 units), CNN filter sizes (32–128), and dropout rates (0.1–0.5) are optimized (using Bayesian hyperparameter optimization using optuna process with $N = 50$ trials) alongside XGBoost parameters to ensure the entire model operates synergistically. This dual-stage optimization ensures that the combination of CNN, LSTM, and XGBoost works together effectively, resulting in a model that addresses the weaknesses of previous architectures and excels in feature representation, temporal pattern capture, and classification accuracy.

The average classification accuracy of the proposed StreXNet model was observed to be 91.60% and 91.81% for two-class and four-class classification, respectively (see Table II), which is superior to other existing methods in the literature. The one-way repeated measure ANOVA (across the classification models) shows a significant difference [$F(4, 68) = 20.261$; $p = 0.000$] in classification accuracy during two-class classification. Furthermore, the pairwise comparison (using Bonferroni correction) found a significantly higher accuracy of the proposed StreXNet model in comparison to other network architectures such as XGBoost ($p = 0.000$), CNN ($p = 0.000$), CNN–LSTM ($p = 0.001$), and XGBoost–CNN ($p = 0.000$). Similarly, one-way repeated measure ANOVA (across the classification models) shows a significant difference [$F(4, 68) = 46.625$; $p = 0.000$] in classification accuracy during four-class classification. Furthermore, the pairwise comparison (using Bonferroni correction) found a significant difference between the proposed StreXNet model and other network architectures, such as XGBoost ($p = 0.000$), CNN ($p = 0.000$), CNN–LSTM ($p = 0.000$), and XGBoost–CNN ($p = 0.000$).

In addition, the one-way repeated measure ANOVA has been performed on precision, recall, and $F1$-score during four-class and two-class classification. A significant difference (across the classification models) has been observed for precision [$F(8136) = 24.703$; $p = 0.000$], recall [$F(8136) = 26.696$; $p = 0.000$], and $F1$-score [$F(8136) = 26.049$; $p = 0.000$] during four-class classification. Furthermore, the pairwise comparison (using Bonferroni correction) found a significantly higher precision ($p = 0.000$), recall ($p = 0.000$), and $F1$-score ($p = 0.000$) of the proposed StreXNet model compared with RF, AdaBoost, SVM, EEGNet, XGBoost, CNN, CNN–LSTM, and XGBoost–CNN. Similarly, during two-class classification, a significant difference (across the classification models) has been observed for precision [$F(8136) = 15.932$; $p = 0.000$], recall [$F(8136) = 27.978$; $p = 0.000$], and $F1$-score [$F(8136) = 23.496$; $p = 0.000$].

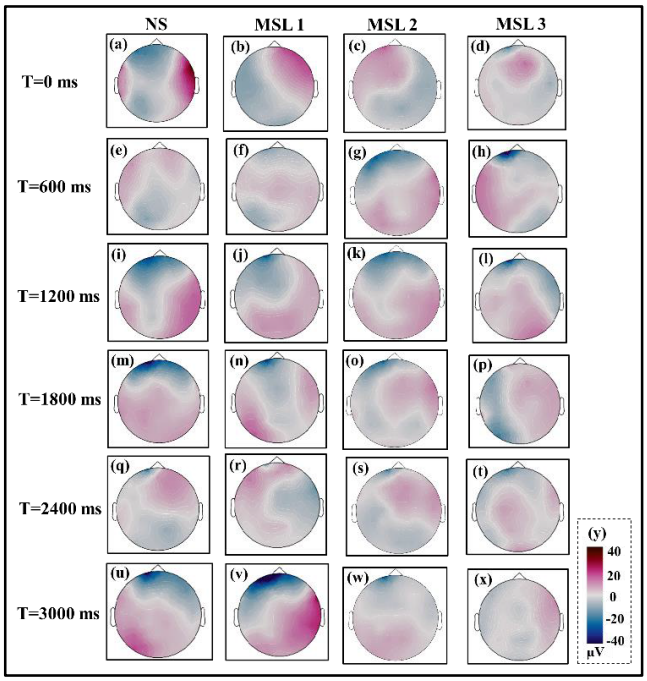


Fig. 6. Representative 2-D topographical brain activity from EEG signals at different time steps for NS, mental stress level 1 (MSL1), mental stress level 2 (MSL2), and mental stress level 3 (MSL3). Here, the higher activation in microvolt (μV) indicates a higher mental stress level compared with low brain activation. (a)–(y) Variation of brain activity at different timesteps across the mental stress levels.

Table III represents the average (SD) of MCC, Jaccard index, and Cohen's kappa during RF, AdaBoost, SVM, EEGNet, XGBoost, CNN, CNN–LSTM, XGBoost–CNN, and the proposed StreXNet model for four- and two-class classifications. The statistical analysis (one-way repeated measure ANOVA) has been conducted for MCC, Jaccard index, and Cohen's kappa. A significant difference (across the classification models) has been observed for MCC [$F(8136) = 8.848$; $p = 0.000$], Jaccard index [$F(8136) = 29.531$; $p = 0.000$], and Cohen's kappa [$F(8136) = 8.838$; $p = 0.000$] during four-class classification. Furthermore, the pairwise comparison (using Bonferroni correction) found a significantly higher MCC ($p < 0.05$), Jaccard index ($p < 0.05$), and Cohen's kappa ($p < 0.05$) of the proposed StreXNet model compared with RF, AdaBoost, SVM, EEGNet, XGBoost, CNN, CNN–LSTM, and XGBoost–CNN. Furthermore, a significant difference has been observed for MCC [$F(8136) = 19.533$; $p = 0.000$], Jaccard index [$F(8136) = 27.319$; $p = 0.000$], and Cohen's kappa [$F(8136) = 19.886$; $p = 0.000$] during two-class classification. The pairwise comparison (using Bonferroni correction) found a significantly higher MCC ($p < 0.05$), Jaccard index ($p < 0.05$), and Cohen's kappa ($p < 0.05$) of the proposed StreXNet model compared with RF, AdaBoost, SVM, EEGNet, XGBoost, CNN, CNN–LSTM, and XGBoost–CNN.

Fig. 6 shows a comparison of representative brain activation across time scales during NS and different stress levels via

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.

TABLE III
MCC, JACCARD INDEX (JI), AND COHEN'S KAPPA AS MENTAL STRESS CLASSIFICATION PERFORMANCE AVERAGED ACROSS SUBJECTS ($N = 18$)

Models	Two-Class Classification			Four-Class Classification		
	MCC (%)	JI (%)	KAPPA (%)	MCC (%)	JI (%)	KAPPA (%)
RF	61.16±10.29	54.50±10.29	59.17±10.47	53.38±12.91	51.27±6.17	51.27±12.83
AdaBoost	53.66±15.58	47.79±15.04	54.23±15.10	46.01±9.54	42.93±5.28	45.01±9.51
SVM	49.55±17.18	43.61±17.91	48.33±17.12	44.76±36.66	41.91±17.77	41.14±36.21
EEGNet	52.90±23.38	53.08±21.11	52.46±23.54	49.42±35.98	50.38±18.53	50.42±35.90
XGBoost	64.17±10.10	59.49±10.90	63.14±9.96	58.74±11.42	64.10±9.32	58.67±11.46
CNN	43.63±20.89	42.98±17.45	43.27±20.10	40.96±24.97	41.49±9.87	41.96±24.94
CNN -LSTM	48.21±20.94	46.33±17.91	48.01±20.51	42.75±27.55	47.91±15.72	43.76±27.42
XGBoost-CNN	72.72±10.01	77.91±1.81	70.60±11.06	68.71±16.42	67.58±9.14	65.69±16.41
Proposed StreXNet	**89.01±5.98**	**85.60±7.68**	**88.97±6.01**	**88.57±8.55**	**84.61±7.44**	**87.98±8.54**

Here, RF, AdaBoost, SVM, EEGNet, XGBoost, CNN, and LSTM represent the random forest, adaptive Boosting, support vector machine, electroencephalography network, extreme gradient boosting, convolutional neural network, and long short-term memory, respectively. MCC, JI, and Kappa indicate Matthew's correlation coefficient. Jaccard Index, and Cohen's Kappa, respectively.

2-D topographical representation. During MSL1, initially, the right frontal lobe was observed to be more activated compared with other brain regions at $T = 0$ ms. In MSL1, at $T = 600$ ms, there is a shift in brain activation toward the frontal-central, parietal, and central region brain regions. Furthermore, at $T = 1200$ ms in MSL1, the higher brain activation location changes to the right temporal, parietal, occipital, and right frontal brain region. Finally, at $T = 2400$ ms, which is the end of MAT question for MSL1, the higher brain activation region shifts to the right frontal and right temporal brain region in the MSL1 condition. However, at $T = 3000$ ms, which is the postquestion state, the frontal brain region (in MSL1 condition) is found to be at a low activation state compared with MSL2 and MSL3 (at $T = 3000$ ms). This could be explained by the fact that at $T = 3000$ ms the MSL2 and MSL3, the participant is engaged in the MAT question; however, for MSL1, the time at $T = 3000$ ms corresponds to the postquestion state [see Fig. 6(v)–(x)] with comparatively less stress as reflected by low activation of the frontal lobe [6]. Similarly, during NS condition, at $T = 0$ ms, $T = 1200$ ms, and $T = 1800$ ms, the frontal brain region is found to be less active indicating a more relaxed state compared with MSL1, MSL2, and MSL3. The proposed StreXNet learns this dynamic variation in the temporal and spatial domains from EEG signals during multilevel mental stress classification.

Table IV represents the comparative study of classification accuracy of the proposed StreXNet model with the previous studies and SOTA models with the present study's dataset. The present study's dataset on the SOTA model of XGBoost [29], ResNet-50) [30], CNN-BWI [30], LSTM) [30], SVM [16], and LDA [31] has achieved an average accuracy of 91.74% and 88.65%, 82.23% and 79.18%, 87.56% and 82.38%, 77.14% and 74.19%, 82.98% and 81.37%, and 82.65% and 78.29% for two-class and four-class classification, respectively (see Table IV). Furthermore, the subject-independent analysis of SDCAN [20] and EA-SDCAN [20] as an SOTA model using the present study's dataset has observed an average accuracy of 69.34% and 65.19%, and 71.28% and 68.57% for two-class and four-class classification, respectively. However, during the four-class classification, the proposed StreXNet model achieved an accuracy of 80.77% and 91.81% for subject-dependent and subject-independent approaches, respectively, representing an improvement in accuracy compared with SOTA models.

IV. DISCUSSION

This study introduces a novel XGBoost integrated deep learning architecture (StreXNet) for end-to-end classification of multilevel mental stress from EEG data. The proposed StreXNet model integrates advanced deep learning techniques with a robust machine learning classifier, combining temporal and spatial feature extraction through a CNN module with SE blocks and an AT-LSTM module. The proposed approach can learn features that are more unique and informative during mental stress tasks than other classification models such as RF, AdaBoost, SVM, EEGNet, XGBoost, CNN, CNN–LSTM, and XGBoost–CNN. In addition, the proposed model demonstrates improvements in four-class mental stress classification performance over other classification architectures (see Tables I and II).

It is important in mental-stress-related studies to minimize the effect of motor engagement while pressing the keys on the keyboard to respond to the questions on a computer screen. To achieve this, the MAT responses in this study were restricted to single-digit numbers ranging from 0 to 9. This approach ensures that the cognitive processes being studied are not influenced by unnecessary motor activity and are consistent across subjects, allowing for a focus on mental activity. The restriction to single-digit responses simplifies the task, minimizing the need for complex motor responses. For multilevel mental stress classification problems, the two methods have been used in previous studies [20], [30]. The first approach focuses on developing a generalized system that can be applied across different subjects performing a similar task, known as subject-independent classification. In the second approach, the classification model is trained on a specific subject and then applied to the same subject for the classification task referred to as subject-dependent classification. In this study, a subject-dependent end-to-end multilevel mental stress classification from EEG data was proposed. Since each subject has a unique way of responding to mental tasks, and the model is trained based on their individual cognitive processes, the subject-dependent approach can achieve even better

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON OF THE PROPOSED CLASSIFICATION FRAMEWORK WITH PREVIOUS WORKS AND SOTA MODELS WITH THE PRESENT STUDY'S DATASET

Study	Number of Subjects	SD / SI Method	Stressor	Acquired Signals	Classifiers	Feature Extraction	Mental States	Accuracy (%)
[29]	26	SD	MAT/SCWT	EEG	XGBoost	Yes	3	90.77
[30]	14	SD	MAT	EEG	ResNet-50	Yes	3	73.44
[30]	14	SD	MAT	EEG	CNN-BWI	Yes	3	90.64
[30]	14	SD	MAT	EEG	LSTM (alpha)	Yes	3	70.67
[16]	22	SD	MAT	EEG	SVM, LR, NB	Yes	4	83.43
[31]	18	SD	MAT	EEG	LDA	Yes	2	84.2
[32]	17	SD	MAT	EEG, ECG, RESP, BVP	RF	Yes	3	84.6
[49]	19 (Male)	SD	MAT	EEG, ECG	Stacking technique	Yes	3	71.57
[20]	21	SI	MAT/Speech	EEG	SDCAN	No	4	58.91
[20]	21	SI	MAT/Speech	EEG	EA-SDCAN	No	4	60.52
Present Study	**18**	SD	MAT	EEG	XGBoost [29]	Yes	2	91.74
							4	88.65
		SD	MAT	EEG	ResNet-50 [30]	Yes	2	82.23
							4	79.18
		SD	MAT	EEG	CNN-BWI [30]	Yes	2	87.56
							4	82.38
		SD	MAT	EEG	LSTM (alpha) [30]	Yes	2	77.14
							4	74.19
		SD	MAT	EEG	SVM [16]	Yes	2	82.98
							4	81.37
		SD	MAT	EEG	LDA [31]	Yes	2	82.65
							4	78.29
		SI	MAT	EEG	SDCAN [20]	No	2	69.34
							4	65.19
		SI	MAT	EEG	EA-SDCAN [20]	No	2	71.28
							4	68.57
		SI	**MAT**	**EEG**	**Proposed StreXNet**	**No**	**2**	**81.93**
							4	**80.77**
		SD	**MAT**	**EEG**		**No**	**2**	**91.60**
							4	**91.81**

Here, SD, SI, MAT, and SCWT represent the subject-dependent, subject-independent, mental arithmetic task, and Stroop color and word test, respectively. EEG, ECG, RESP, and BVP indicate electroencephalography, electrocardiography, respiration rate, and blood volume pulse, respectively. XGBoost, ResNet-50, CNN-BWI, LSTM, SVM, LR, NB, LDA, RF, SDCAN, and EA-SDCAN represent the extreme gradient boosting, residual network with 50 layers, convolutional neural network-brain wave images, long short-term memory, support vector machine, logistic regression, naive Bayes, linear discriminant analysis, random forest, symmetric deep convolutional adversarial network, and euclidean space data alignment approach based SDCAN respectively.

classification performance, provided that enough classification data are recorded.

The comparative study of the classification accuracy of the proposed StreXNet model with previous studies during multilevel mental stress classification is presented (see Table IV). The previous studies have achieved an average accuracy of 90.77% (XGBoost) [29], 73.44% (ResNet-50) [30], 90.64% (CNN-BWI) [30], 70.67% (LSTM) [30], 83.43% (SVM, LR, NB) [16], 84.2% (LDA) [31], 84.6% (RF) [32], and 71.57% (stacking technique) [49] (see Table IV). However, these studies have used manual feature extraction during the classification process. The previous study has also used an end-to-end multilevel mental stress classification (subject independent) with an average accuracy of 58.91% (SDCAN) [20] and 60.52% (EA-SDCAN) [20]. The present study has compared the SOTA model (using the current study's dataset) with the proposed StreXNet model, as shown in Table IV. The proposed StreXNet model demonstrates improved accuracy compared with previous studies.

The CM for two-class classification [see Fig. 5(a)] shows that out of 924 actual "S" instances, the model correctly predicted 884 instances as "S" and misclassified 40 as "NS." Similarly, for "NS," 193 out of 231 instances were correctly predicted, but 38 were misclassified as "S." The proposed model shows superior performance, correctly identifying 95.67% and 83.5% of "S" and "NS" cases, respectively. Similarly, for multiclass classification, the matrix shows a much more distinct separation among the four classes (NS, MSL1, MSL2, and MSL3). In total, 1155 instances, 231 for NS, and 308 each for MSL1, MSL2, and MLS3. The maximum misclassification of 4.33% (for NS condition) was found between NS and MSL1 during multiclass classification [CM in Fig. 5(b)].

This appears obvious since MSL1 involved one-digit calculations and, therefore, not being able to create enough mental stress reflected in EEG to be distinguished from EEG without stress. Both the conditions (NS and MLS1) may produce similar levels of cognitive engagement, leading to overlapping features that the model struggles to distinguish. In addition, the four-class classification accuracy has been found to be almost similar compared with the two-class classification from the proposed StreXNet model. It could be due to more granular differentiation of mental stress levels, capturing subtle distinctions in EEG signals that might be missed in a two-class system. This allows the model to recognize and correctly classify a wider range of physiological responses associated with different levels of stress.

The broader implications of our findings could impact real-life personalized stress detection applications, particularly when the algorithm is integrated into portable stress

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.

monitoring devices. The robustness of mental stress classification from EEG data offers a scalable framework that could be used in mental health monitoring and brain–computer interface systems from EEG sensors. Furthermore, the proposed EEG classification method has the potential to be applied in other areas, such as classifying different stages of Alzheimer's, Parkinson's, and other neurodegenerative diseases (using preprocessed EEG data) in individuals by analyzing neurological brain activity through time-series brain activation.

V. Limitations and Future Scope

Despite the promising outcomes, there are several limitations of the present study. First, we only chose male volunteers to rule out the potential for the hormone cycle to alter the stress induction process. In addition, the previous study indicates that women are more likely to develop stress disorders than men, even when exposed to similar types of stimuli [50]. Second, only one session for each level was conducted in this study. Higher accuracy could have been achieved by having more trials to get more labeled data. However, unrest could be observed in participants as nearly 2.5 h were used for one trial, including 1 h of instrumentation setup (electrode placement). More than one trial for each level could have taken another 45 min (including a rest period between sessions), which was inconvenient for the volunteers. Third, the current study involves university-educated individuals with similar MMSE scores. Future studies will focus on whether our findings of mental stress classification can be generalized to both genders. In addition, future research will include a broader range of MMSE scores and different mental tasks during mental stress classification. Furthermore, future work will explore the extension of the present study with larger and more varied clinical populations.

VI. Conclusion

This research presents a robust and end-to-end approach for multilevel mental stress identification from a novel architecture named StreXNet. The proposed model, which uses EEG and integrates SE block enhanced CNN, AT-LSTM module, and XGBoost classifier, demonstrates improved performance in classifying mental stress levels compared with other classification models such as RF, AdaBoost, SVM, EEGNet, XGBoost, CNN, CNN–LSTM, and XGBoost–CNN. The proposed StreXNet model has achieved an average classification of 91.81% and 80.77% for four-class classification during subject-dependent and subject-independent approaches, respectively. Furthermore, a comparison of the proposed StreXNet architecture with previous works and SOTA models highlighted its effectiveness and reliability in EEG-sensor-based mental stress classification. Furthermore, one-way repeated measure ANOVA (across the classification models) shows a significant difference [$F(4, 68) = 46.625$; $p = 0.000$] in classification accuracy during four-class classification. The pairwise comparison (using Bonferroni correction) found a significant difference between the proposed StreXNet model and other network architectures, such as XGBoost ($p = 0.000$), CNN ($p = 0.000$), CNN–LSTM ($p = 0.000$), and XGBoost–CNN ($p = 0.000$). The potential of StreXNet extends beyond robust and efficient classification, offering a scalable and adaptable framework for real-world personalized mental stress detection, especially in portable mental stress monitoring systems. Future work will focus on expanding the dataset to include a more diverse clinical population, enabling subject-independent studies for real-time mental stress classification. In addition, further exploration into reducing the number of EEG channels while maintaining high accuracy could enhance the model's practicality in real-world applications. These advancements will contribute to the development of more precise, reliable, and widely applicable EEG-sensor-based mental stress identification systems.

Data and Codes Are Available: https://github.com/NRL-IITD/StreXNet

Acknowledgment

The authors would like to express their gratitude to the members of the IITD Neuromechanics Lab and the volunteer participants for their assistance during the rigorous data acquisition procedure.

References

[1] K. J. Lovelace, C. C. Manz, and J. C. Alves, "Work stress and leadership development: The role of self-leadership, shared leadership, physical fitness and flow in managing demands and increasing job control," *Hum. Resource Manage. Rev.*, vol. 17, no. 4, pp. 374–387, Dec. 2007, doi: 10.1016/j.hrmr.2007.08.001.

[2] H. A. Khan, T. N. Nguyen, G. Shafiq, J. Mirza, and M. A. Javed, "A secure wearable framework for stress detection in patients affected by communicable diseases," *IEEE Sensors J.*, vol. 23, no. 2, pp. 981–988, Jan. 2023, doi: 10.1109/JSEN.2022.3204586.

[3] R. S. Lazarus, "From psychological stress to the emotions: A history of changing outlooks," *Annu. Rev. Psychol.*, vol. 44, no. 1, pp. 1–21, Jan. 1993, doi: 10.1146/annurev.psych.44.1.1.

[4] G. Fink, "Stress: Definition and history," *Stress Sci. Neuroendocrinol.*, vol. 3, no. 9, pp. 3–14, 2010, doi: 10.1016/B978-008045046-9.00076-0.

[5] N. Fekadu, W. Shibeshi, and E. Engidawork, "Major depressive disorder: Pathophysiology and clinical management," *J. Depression Anxiety*, vol. 6, no. 1, pp. 255–257, 2017, doi: 10.4172/2167-1044.1000255.

[6] S. Evans-Lacko et al., "Socio-economic variations in the mental health treatment gap for people with anxiety, mood, and substance use disorders: Results from the WHO world mental health (WMH) surveys," *Psychol. Med.*, vol. 48, no. 9, pp. 1560–1571, Jul. 2018, doi: 10.1017/s0033291717003336.

[7] E.-H. Lee, "Review of the psychometric evidence of the perceived stress scale," *Asian Nursing Res.*, vol. 6, no. 4, pp. 121–127, Dec. 2012, doi: 10.1016/j.anr.2012.08.004.

[8] K. B. Koh, J. K. Park, C. H. Kim, and S. Cho, "Development of the stress response inventory and its application in clinical practice," *Psychosomatic Med.*, vol. 63, no. 4, pp. 668–678, Jul. 2001, doi: 10.1097/00006842-200107000-00020.

[9] J. E. Dise-Lewis, "The life events and coping inventory: An assessment of stress in children," *Psychosomatic Med.*, vol. 50, no. 5, pp. 484–499, Sep. 1988, doi: 10.1097/00006842-198809000-00005.

[10] C. D. Spielberger, F. Gonzalez-Reigosa, A. Martinez-Urrutia, L. Natalicio, and D. S. Natalicio, "Development of the Spanish edition of the state-trait anxiety inventory," *Interam. J. Psychol.*, vol. 5, nos. 3–4, pp. 145–158, 1971, doi: 10.4135/9781483365817.n1316.

[11] W. Seo, N. Kim, C. Park, and S.-M. Park, "Deep learning approach for detecting work-related stress using multimodal signals," *IEEE Sensors J.*, vol. 22, no. 12, pp. 11892–11902, Jun. 2022, doi: 10.1109/JSEN.2022.3170915.

[12] K. Dedovic, R. Renwick, N. K. Mahani, V. Engert, S. J. Lupien, and J. C. Pruessner, "The Montreal imaging stress task: Using functional imaging to investigate the effects of perceiving and processing psychosocial stress in the human brain," *J. Psychiatry Neurosci.*, vol. 30, no. 5, pp. 319–325, 2005.

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.

[13] G. Qu et al., "A hybrid critical channel selection framework for EEG emotion recognition," *IEEE Sensors J.*, vol. 24, no. 9, pp. 14881–14893, May 2024, doi: 10.1109/JSEN.2024.3380749.

[14] Y. Zheng, T. C. H. Wong, B. H. K. Leung, and C. C. Y. Poon, "Unobtrusive and multimodal wearable sensing to quantify anxiety," *IEEE Sensors J.*, vol. 16, no. 10, pp. 3689–3696, May 2016, doi: 10.1109/JSEN.2016.2539383.

[15] Y. Zhang, Q. Wang, Z. Y. Chin, and K. K. Ang, "Investigating different stress-relief methods using electroencephalogram (EEG)," in *Proc. 42nd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2020, pp. 2999–3002, doi: 10.1109/EMBC44109.2020.9175900.

[16] A. R. Subhani, W. Mumtaz, M. N. B. M. Saad, N. Kamel, and A. S. Malik, "Machine learning framework for the detection of mental stress at multiple levels," *IEEE Access*, vol. 5, pp. 13545–13556, 2017, doi: 10.1109/ACCESS.2017.2723622.

[17] G. Giannakakis, D. Grigoriadis, K. Giannakaki, O. Simantiraki, A. Roniotis, and M. Tsiknakis, "Review on psychological stress detection using biosignals," *IEEE Trans. Affect. Comput.*, vol. 13, no. 1, pp. 440–460, Jan. 2022, doi: 10.1109/TAFFC.2019.2927337.

[18] E. S. Epel et al., "More than a feeling: A unified view of stress measurement for population science," *Frontiers Neuroendocrinol.*, vol. 49, pp. 146–169, Apr. 2018, doi: 10.1016/j.yfrne.2018.03.001.

[19] S. S. Dickerson and M. E. Kemeny, "Acute stressors and cortisol responses: A theoretical integration and synthesis of laboratory research," *Psychol. Bull.*, vol. 130, no. 3, pp. 355–391, 2004, doi: 10.1037/0033-2909.130.3.355.

[20] R. Fu et al., "Symmetric convolutional and adversarial neural network enables improved mental stress classification from EEG," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 30, pp. 1384–1400, 2022, doi: 10.1109/TNSRE.2022.3174821.

[21] N. Z. Gurel, H. Jung, S. Hersek, and O. T. Inan, "Fusing near-infrared spectroscopy with wearable hemodynamic measurements improves classification of mental stress," *IEEE Sensors J.*, vol. 19, no. 19, pp. 8522–8531, Oct. 2019, doi: 10.1109/JSEN.2018.2872651.

[22] A. Hag, D. Handayani, T. Pillai, T. Mantoro, M. H. Kit, and F. Al-Shargie, "EEG mental stress assessment using hybrid multidomain feature sets of functional connectivity network and time-frequency features," *Sensors*, vol. 21, no. 18, p. 6300, Sep. 2021, doi: 10.3390/s21186300.

[23] T. B. Tang, L. W. Yeo, and D. J. H. Lau, "Activity awareness can improve continuous stress detection in galvanic skin response," in *Proc. IEEE SENSORS*, Nov. 2014, pp. 1980–1983, doi: 10.1109/ICSENS.2014.6985421.

[24] P. Karthikeyan, M. Murugappan, and S. Yaacob, "Analysis of stroop color word test-based human stress detection using electrocardiography and heart rate variability signals," *Arabian J. Sci. Eng.*, vol. 39, no. 3, pp. 1835–1847, Mar. 2014, doi: 10.1007/s13369-013-0786-8.

[25] O. M. Mozos et al., "Stress detection using wearable physiological and sociometric sensors," *Int. J. Neural Syst.*, vol. 27, no. 2, Mar. 2017, Art. no. 1650041, doi: 10.1142/s0129065716500416.

[26] J. K. Meier, M. Weymar, and L. Schwabe, "Stress alters the neural context for building new memories," *J. Cognit. Neurosci.*, vol. 32, no. 12, pp. 2226–2240, Dec. 2020, doi: 10.1162/jocn_a_01613.

[27] D. S. Krantz, H. T. Santiago, W. J. Kop, C. N. B. Merz, A. Rozanski, and J. S. Gottdiener, "Prognostic value of mental stress testing in coronary artery disease," *Amer. J. Cardiol.*, vol. 84, no. 11, pp. 1292–1297, Dec. 1999, doi: 10.1016/S0002-9149(99)00560-3.

[28] S. S. Panicker and P. Gayathri, "A survey of machine learning techniques in physiology based mental stress detection systems," *Biocybernetics Biomed. Eng.*, vol. 39, no. 2, pp. 444–469, Apr. 2019, doi: 10.1016/j.bbe.2019.01.004.

[29] Y.-H. Tsai, S.-K. Wu, S.-S. Yu, and M.-H. Tsai, "Analyzing brain waves of table tennis players with machine learning for stress classification," *Appl. Sci.*, vol. 12, no. 16, p. 8052, Aug. 2022, doi: 10.3390/app12168052.

[30] M. A. Hafeez and S. Shakil, "EEG-based stress identification and classification using deep learning," *Multimedia Tools Appl.*, vol. 83, no. 14, pp. 42703–42719, Oct. 2023, doi: 10.1007/s11042-023-17111-0.

[31] M. Bagheri and S. D. Power, "Investigating hierarchical and ensemble classification approaches to mitigate the negative effect of varying stress state on EEG-based detection of mental workload level—And vice versa," *Brain-Comput. Interfaces*, vol. 8, nos. 1–2, pp. 26–37, Apr. 2021, doi: 10.1080/2326263x.2021.1948756.

[32] M. Zanetti et al., "Multilevel assessment of mental stress via network physiology paradigm using consumer wearable devices," *J. Ambient Intell. Humanized Comput.*, vol. 12, no. 4, pp. 4409–4418, 2021, doi: 10.1007/s12652-019-01571-0.

[33] J. C. Pruessner et al., "Deactivation of the limbic system during acute psychosocial stress: Evidence from positron emission tomography and functional magnetic resonance imaging studies," *Biol. Psychiatry*, vol. 63, no. 2, pp. 234–240, Jan. 2008, doi: 10.1016/j.biopsych.2007.04.041.

[34] A. Kumar, R. Bajpai, and D. Joshi, "Temperature profile of nasal breathing for respiration monitoring in ambulatory settings," *IEEE Sensors J.*, vol. 22, no. 23, pp. 23328–23340, Dec. 2022, doi: 10.1109/JSEN.2022.3212920.

[35] F. Al-shargie, T. B. Tang, N. Badruddin, and M. Kiguchi, "Towards multilevel mental stress assessment using SVM with ECOC: An EEG approach," *Med. Biol. Eng. Comput.*, vol. 56, no. 1, pp. 125–136, Jan. 2018, doi: 10.1007/s11517-017-1733-8.

[36] E. Alyan, N. M. Saad, and N. Kamel, "Effects of workstation type on mental stress: FNIRS study," *Hum. Factors*, vol. 63, no. 7, pp. 1230–1255, Nov. 2021, doi: 10.1177/0018720820913173.

[37] A. Ciaramidaro, J. Toppi, C. Casper, C. M. Freitag, M. Siniatchkin, and L. Astolfi, "Multiple-brain connectivity during third party punishment: An EEG hyperscanning study," *Sci. Rep.*, vol. 8, no. 1, pp. 1–13, May 2018, doi: 10.1038/s41598-018-24416-w.

[38] X. C. E. Vrijdag, H. van Waart, R. M. Pullon, C. Sames, S. J. Mitchell, and J. W. Sleigh, "EEG functional connectivity is sensitive for nitrogen narcosis at 608 kPa," *Sci. Rep.*, vol. 12, no. 1, p. 4880, Mar. 2022, doi: 10.1038/s41598-022-08869-8.

[39] A. Khajuria and D. Joshi, "EEG-explained cortical correlates of transfemoral amputees during balancing with vibrotactile feedback: A pilot study," *Med. Eng. Phys.*, vol. 101, Mar. 2022, Art. no. 103772, doi: 10.1016/j.medengphy.2022.103772.

[40] M. M. Artĕncio, J. D. M. E. Giraldi, and J. H. C. de Oliveira, "A cup of black coffee with GI, please! Evidence of geographical indication influence on a coffee tasting experiment," *Physiol. Behav.*, vol. 245, Mar. 2022, Art. no. 113671, doi: 10.1016/j.physbeh.2021.113671.

[41] I. Al-Shourbaji et al., "A deep batch normalized convolution approach for improving COVID-19 detection from chest X-ray images," *Pathogens*, vol. 12, no. 1, p. 17, Dec. 2022, doi: 10.3390/pathogens12010017.

[42] X. Jin, Y. Xie, X.-S. Wei, B.-R. Zhao, Z.-M. Chen, and X. Tan, "Delving deep into spatial pooling for squeeze-and-excitation networks," *Pattern Recognit.*, vol. 121, Jan. 2022, Art. no. 108159, doi: 10.1016/j.patcog.2021.108159.

[43] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017, doi: 10.1145/3065386.

[44] M. M. Khan, M. S. Uddin, M. Z. Parvez, and L. Nahar, "A squeeze and excitation ResNeXt-based deep learning model for Bangla handwritten compound character recognition," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 34, no. 6, pp. 3356–3364, Jun. 2022, doi: 10.1016/j.jksuci.2021.01.021.

[45] X. Zheng and W. Chen, "An attention-based bi-LSTM method for visual object classification via EEG," *Biomed. Signal Process. Control*, vol. 63, Jan. 2021, Art. no. 102174, doi: 10.1016/j.bspc.2020.102174.

[46] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014.

[47] J. Shi et al., "Control parameter extraction model for reconfigurable bandpass filter based on Bayesian optimized multiple-output XGBoost," *IEEE Microw. Wireless Technol. Lett.*, vol. 34, no. 3, pp. 263–266, Mar. 2024, doi: 10.1109/LMWT.2023.3343095.

[48] F. Miao, L. Yao, and X. Zhao, "Adaptive margin aware complement-cross entropy loss for improving class imbalance in multi-view sleep staging based on EEG signals," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 30, pp. 2927–2938, 2022, doi: 10.1109/TNSRE.2022.3213848.

[49] A. Hemakom, D. Atiwiwat, and P. Israsena, "ECG and EEG based detection and multilevel classification of stress using machine learning for specified genders: A preliminary study," *PLoS ONE*, vol. 18, no. 9, Sep. 2023, Art. no. e0291070, doi: 10.1371/journal.pone.0291070.

[50] A. Othmani, B. Brahem, and Y. Haddou, "Machine-learning-based approaches for post-traumatic stress disorder diagnosis using video and EEG sensors: A review," *IEEE Sensors J.*, vol. 23, no. 20, pp. 24135–24151, Oct. 2023, doi: 10.1109/JSEN.2023.3312172.

Authorized licensed use limited to: COLLEGE OF ENGINEERING - Pune. Downloaded on March 23,2026 at 08:56:49 UTC from IEEE Xplore. Restrictions apply.